

McGill University
ECSE 222 – Digital Logic (Fall 2025)
Pen-and-Paper Assignment #1

Instructions

- TA in charge: Sihui Shen (sihui.shen@mail.mcgill.ca)
- Write/type your answers in the space provided. Don't provide logic circuits from simulation software.
- Due date: Thursday 25th of September 2025 at 11:59 PM.
- You must electronically submit the assignment in the form of a PDF file (only this format will be accepted) on myCourses before the deadline. You may submit as many times as you want, however, only the last submission will be kept by the system.
- You must solve the assignment in teams of two/three students, whereas only one of the students should perform the actual electronic submission.
- Late submissions will incur penalties as described in the course syllabus.

Student identification

First student full name	
First student ID number	
Second student full name	
Second student ID number	
Third student full name (if applicable)	
Third student ID number (if applicable)	

Evaluation (for course graders use only)

Question 1 (15%)	Question 2 (20%)	Question 3 (15%)	Question 4 (20%)	Total
Question 5 (05%)	Question 6 (15%)	Question 7 (10%)		

Question 1 (15%)

Consider the following logic expression. Write down the truth table for the expression

$$F = (X + \bar{X}Y)(\bar{Y} + X\bar{Z})$$

X	Y	Z	F	
0	0	0	0	$(0+1\cdot0)\cdot(1+0\cdot1)$
0	0	1	0	$(0+1\cdot0)\cdot(1+0\cdot0)$
0	1	0	0	$(0+1\cdot1)\cdot(0+0\cdot1)$
0	1	1	0	$(0+1\cdot1)\cdot(0+0\cdot0)$
1	0	0	1	$(1+0\cdot0)\cdot(1+1\cdot0)$
1	0	1	1	$(1+0\cdot0)\cdot(1+1\cdot1)$
1	1	0	1	$(1+0\cdot1)\cdot(0+1\cdot1)$
1	1	1	0	$(1+0\cdot1)\cdot(0+1\cdot0)$

Question 2 (20%)

Consider the following logic expression. Write down the VHDL code for both structural programming and behavioral programming

$$F = (X + \bar{X}Y)(\bar{Y} + X\bar{Z})$$

Structural Pattern (10%)

```
library IEEE;
use IEEE.std_logic_1164.all;

entity func is
    port ( X, Y, Z : IN std_logic;
          F : OUT std_logic);
end func;

architecture structural of func is
begin
    F <= (X OR (NOT(X) AND Y))
        AND (NOT(Y) OR (X AND NOT(Z)))
end structural
```

Behavioral Pattern (10%)

```
library IEEE;
use IEEE.std_logic_1164.all;

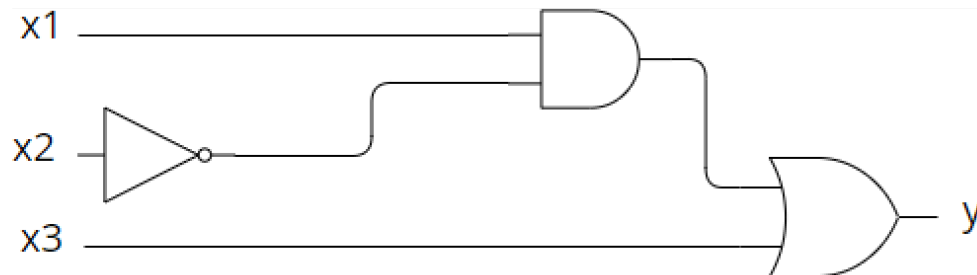
entity func is
    port ( X, Y, Z : IN std_logic;
          F : OUT std_logic);
end func;

architecture behavioral of func is
    signal s :
        std_logic_vector(2 downto 0);
begin
    s <= X & Y & Z;

    with s select
        F <= '1' when
            '100' | '101' | '110',
            '0' when others;
end behavioral
```

Question 3 (15%)

Consider the following logic circuit. In the space below, provide the corresponding truth table and logic function.



Truth table (10%)	Logic function (05%)

Question 4 (20%)

Consider the following truth table.

x	y	z	$f(x,y,z)$
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

a) From the truth table, provide the canonical sum-of-products (SOP) expression (5%)

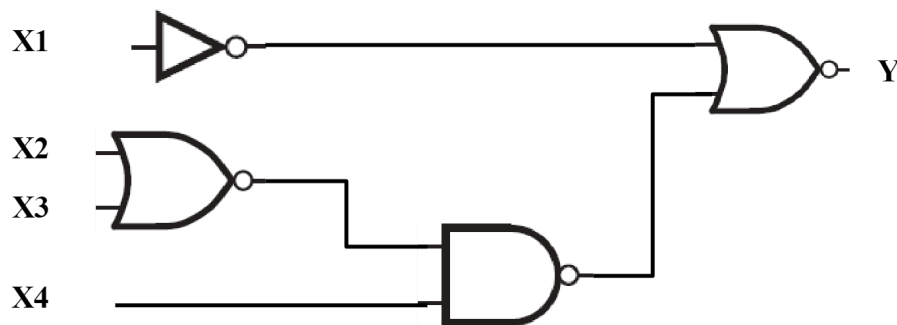
b) Using Boolean algebra show that the canonical SOP expression can be simplified to $y + \bar{x}\bar{z}$ (5%)

c) From the truth table, provide the canonical product-of-sums (POS) expression (05%)

d) Using Boolean algebra, simplify the canonical POS expression and show that it can be implemented by a logic circuit of cost 6. Assume that the complements of the input variables are provided at no additional cost. Draw the circuit (05%)

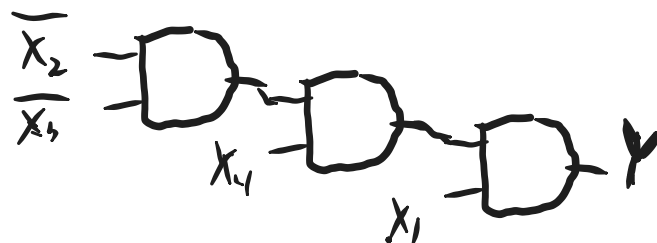
Question 5 (05%)

Consider the following circuit.



Re-Synthesize the above circuit using only AND gates. Assume that the complements of the input variables are provided at no additional cost.

$$\begin{aligned}
 Y &= \left(\overline{X_2 + X_3} \right) X_4 + \bar{X}_1 \\
 &= \left(\left(\overline{X_2 + X_3} \right) X_4 \right) X_1 \quad \text{De Morgan's} \\
 Y &= \left(\left(\overline{X_2 \cdot \bar{X}_3} \right) X_4 \right) X_1 \quad \text{De Morgan's}
 \end{aligned}$$



Question 6 (15%)

Consider the following logic expression

$$F = \bar{x}\bar{z} + x\bar{y} + xz$$

Synthesize the above expression using only NOR gates. Assume that the complements of the input variables are provided at no additional cost.

Question 7 (10%)

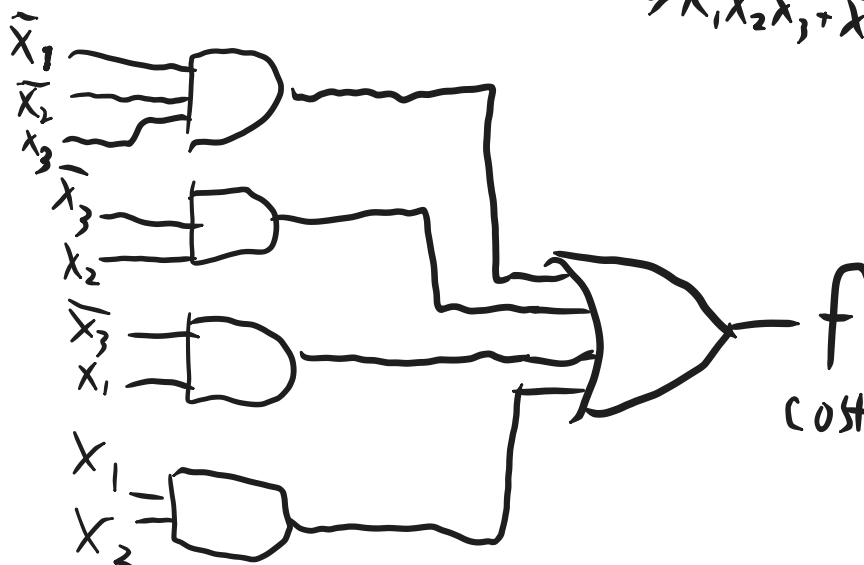
Consider the following table.

X1	X2	X3	f
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

$$\begin{aligned}
 &X_1 + X_2 + X_3 \\
 &\bar{X}_1 \bar{X}_2 X_3 \\
 &\bar{X}_1 X_2 \bar{X}_3 \\
 &X_1 + \bar{X}_2 + \bar{X}_3 \\
 &\bar{X}_1 + X_2 + \bar{X}_3 \\
 &X_1 \bar{X}_2 \bar{X}_3 \\
 &X_1 X_2 \bar{X}_3 \\
 &X_1 X_2 X_3
 \end{aligned}$$

a) Provide the expression of sum-of-products and draw the minimal circuit. Provide the cost of this logic circuit. Assume that the complements of the input variables are provided at no additional cost. (5%)

$$\begin{aligned}
 f &= \bar{X}_1 \bar{X}_2 X_3 + \bar{X}_1 X_2 \bar{X}_3 + X_1 \bar{X}_2 \bar{X}_3 + X_1 X_2 \bar{X}_3 + X_1 X_2 X_3 \\
 &= \bar{X}_1 \bar{X}_2 X_3 + X_1 \bar{X}_2 \bar{X}_3 + X_1 X_2 \bar{X}_3 + X_2 \bar{X}_3 \\
 &= \bar{X}_1 \bar{X}_2 X_3 + X_1 \bar{X}_2 \bar{X}_3 + X_2 (\bar{X}_1 X_3 + \bar{X}_3) \\
 &= \bar{X}_1 \bar{X}_2 X_3 + X_1 \bar{X}_2 \bar{X}_3 + X_2 (\bar{X}_3 + X_1) \\
 &= \bar{X}_1 \bar{X}_2 X_3 + X_1 \bar{X}_2 \bar{X}_3 + X_2 \bar{X}_3 + X_1 X_2 \\
 &\Rightarrow \bar{X}_1 \bar{X}_2 X_3 + \bar{X}_3 X_2 + \bar{X}_3 X_1 + X_1 X_2
 \end{aligned}$$



$$\begin{aligned}
 \text{cost} &= 5 \text{ gates} \\
 &+ 13 \text{ inputs} \\
 &= 18
 \end{aligned}$$

b) Provide the expression of product-of-sums and draw the minimal circuit. Provide the cost of this logic circuit. Assume that the complements of the input variables are provided at no additional cost. (5%)

$$f = (x_1 + x_2 + x_3)(x_1 + \bar{x}_2 + \bar{x}_3)(\bar{x}_1 + x_2 + \bar{x}_3)$$

Can't be reduced further in POS form

